

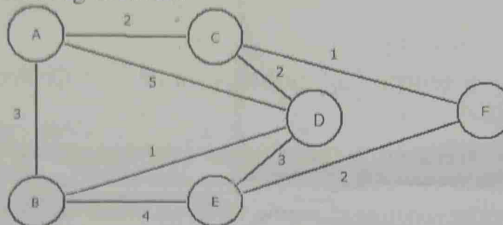


Answer any five questions of the followings.

$5 \times 8 = 40$

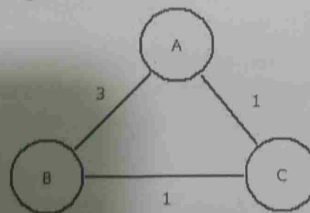
1. (a) Define routing algorithm with its classification. 2
(b) Write the basic steps for a router to find the shortest path. 2
(c) List the advantages and disadvantages of static and dynamic routing algorithm. 4

2. (a) Write the routing steps for link-state routing algorithm. 3
(b) Consider the following network: 5



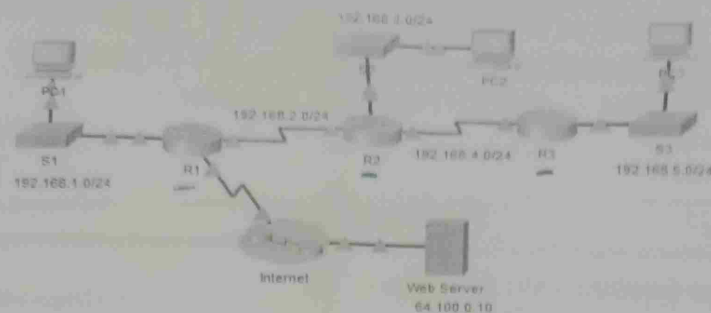
Find the shortest path using the Dijkstra's Algorithm.

3. (a) What is Vulnerable Time in carrier sense multiple access? 2
(b) Write the routing algorithm of Distance Vector. 3
(c) Consider the following diagram: 3



Calculate the shortest path using Distance Vector routing algorithm.

4. (a) Write the routing protocol messages of Open Shortest Path First (OSPF). 3
(b) Write the generic link state routing processes to reach a state of convergence of OSPF routing protocol. 5
5. (a) Define flush time. Write the characteristics of Routing Information Protocol (RIP). 3
(b) Consider the following networks: 5



Write the appropriate commands for configuring the RIPv2 of router 1, 2 and 3.

6. (a) Define namespace of Domain Name System (DNS). Briefly explain the subdomain of DNS protocol. 3
- (b) Explain the top level domain of DNS server with example. 5
7. (a) Write the IP range of private network addresses. 2
- (b) Explain the error-reporting messages of Internet Control Message Protocol (ICMP). 3
- (c) Consider the scenario: in class less inter-domain routing the IP addresses in a corporate network are obtained from the service provider. Changing the service provider requires changing all IP addresses in the network. Now briefly explain how Network Address Translator (NAT) supporting migration between the network service providers? 3



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE311: Computer Architecture

Final Examination

Semester: Fall 2024

Batch: 23rd

Total marks: 40

Date: 22/03/2025

Time: 2 Hours

Answer any five questions of the followings.

5 × 8 = 40

1. a) Discuss the significance of the transition from Motorola 680X0 to PowerPC in the evolution of the Apple Macintosh. 4
b) Draw the organization of a typical personal computer from the mid-1990s. 4
2. a) Explain the role of cache memory as a performance optimization technique in computer architecture. 2
b) A CPU's speed is 3.7 GHz, the number of instructions executed is 5×10^9 , and the average cycle per instruction is 3. Calculate the CPU execution time. 3
c) Use the necessary information from the question '2b', calculate how many millions of instructions the CPU executes per second. 3
3. a) In an accumulator-based CPU that supports only single-address instructions, the Accumulator (AC) and Data Register (DR) can serve as implicit operands. How do AC and DR act here as implicit operands? 3
b) How does an accumulator-based CPU with single-address instructions implement multi-operand operations like $Z := X + Y$? Provide a step-by-step explanation. 5
4. a) Explain the structure of the Current Program Status Register (CPSR) and the role of its condition flags in program execution. 2
b) Given the initial values $R1 := 4$ and $R0 := 10$, determine the value of R0 register after the execution of the following instruction. 3

MOV R0, R1, LSL #2

c) State the advantages of Memory-Mapped I/O in computer systems and explain how it improves the system design and performance. 3
5. a) Define microcomputer interfacing and state its importance. 2
b) What is the role of microcode in microprogrammed control? 2
c) How do support chips improve the performance of microcomputer systems? 4
6. a) What is redundancy in fault-tolerant systems? 2
b) Write the key characteristics of instruction execution in fault-tolerant systems. 3
c) State the differences between fixed-point and floating-point number formats. 3
7. Given the memory access sequence 3, 2, 2, 1, 3, 4, 2, 1, 7, 9, 0, 5, 6, 3, 1, 3, 5, 8, 7, 1, 1, 3, 5 and a cache size of 3, determine the most suitable cache replacement algorithm among LRU, FIFO, and LFU. Justify your choice by simulating cache behavior and comparing hit/miss rates for each algorithm. 8



HAMDARD UNIVERSITY BANGLADESH

Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT317
Final Examination
Date: 3/24/2025

Course Title: Engineering Mathematics
Semester: Fall 2024
Total Marks: 40

Total Time: 2 Hours

Answer any four (04) out of the following questions:

$4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

- | | Marks |
|--|-------|
| Q1. (a) Define the analytic and singular function. | [2] |
| (b) Using power series solve the following differential equation | [8] |
| i. $y'' + xy' + y = 0$. | |
| ii. $\frac{dy}{dx} - y = 0$. | |
| Q2. (a) Solve the initial value problem $\frac{dy}{dt} - 2y = e^{5t}$ where, $y(0) = 3$ | [5] |
| (b) Find the inverse Laplace of the following statement | [5] |
| i. $\frac{1}{s^3 + 4s^2 + 3s}$ | |
| ii. $\frac{1}{s^3 + 2s}$ | |
| Q3. (a) State and prove the generating function of Laguerre Polynomial. | [5] |
| (b) Express $f(x) = x^3 - 6x^2 + 7x + 9$ in terms of Laguerre polynomial. | [5] |
| Q4. (a) State and prove the Parseval's Formula. | [4] |
| (b) Determine the Fourier series of the function $f(x) = \begin{cases} 0, & -\pi \leq x \leq 0 \\ x, & 0 < x \leq \pi \end{cases}$ | [6] |
| Hence, evaluate the sum $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$ | |
| Q5. (a) Generate the trigonometric representation of Bessel Function. | [4] |
| (b) Prove that $J_0(x) = \frac{2}{\pi} \int_0^1 \frac{\cos xt}{\sqrt{1-t^2}} dt$. | [3] |
| (c) Find the complex form of Fourier series. | [3] |
| Q6. (a) State and prove the generating function of Bessel function. | [6] |
| (b) Prove that, $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$. | [4] |

$$\frac{1}{3} e^{4t} + \frac{8}{2} e^{2t}$$

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 313: Database Management System

Batch: 23rd

Semester: Fall 2024

Date: 16/03/2025

Final Examination

Time: 2 hours

Total Marks: 40

Answer any five questions.

Marks: 5 x 8 = 40

1. (a) State the name of two main collections of relational operation. Write the SQL to determine the Union, Intersection and Subtraction operation for the following relation A and B. 4

First	Last	Age
Bill	Smith	22
Sally	Green	28
Mary	Keen	23
Tony	Jones	32

First	Last	Age
Forest	Gump	36
Sally	Green	28
Donja	De Marco	27

- (b) Write the SQL query and Relational Algebra to determine different types of outer-join in DBMS. 4
2. (a) How index are created in DBMS? 2
- (b) Define index in DBMS. Write the syntax to create and remove an index in DBMS. 3
- (c) State the guidelines for creating an Index in DBMS. 3
3. (a) Write the SQL query to create a sequence named **dept_deptid_seq** which is start with 100, increment by 10 and the max value is 1000. 2
- (b) Draw the syntax to insert and modify a sequence in DBMS. 3
- (c) What is meant by a sequence in DBMS? Write the guidelines for modifying a sequence. 3
4. (a) Write the differences between **join** and **natural join** in DBMS. 2
- (b) Consider the following Student table 6

Std_id	Std_name	Std_email	Std_contact	Std_addr	Std_bg	Std_dept	ob_mrk
315121003	Afia Khan	a@gmail.com	01913000000	Dhaka	A +ve	CSE	72
315121004	Rabeya	b@yahoo.com	01721111111	Sonargaon	O +ve	CSE	66
555121001	Siam	c@hub.edu.bd	01821222222	Dinajpur	B +ve	EEE	56
536142004	Sadia	d@gmail.com	01911333333	Narayanganj	O +ve	Econ	52
222121003	Siam	e@hub.edu.bd	01551444444	Munshigonj	O -ve	BBA	56

- i) Using SQL, create a table Std_details (std_id, std_name, std_email, std_contact, std_bg, ob_mark) from the Student table.
- ii) Derive the SQL to insert Siam's std dob = '05-05-1995' who is in EEE in the table std_details.
- iii) Using the table Student, Write the SQL to create a view Std_vu (std_id, std_name, std_email, std_contact, std_addr, std_bg) for CSE department.
5. (a) Compare among different types of data fragmentation. 4
- (b) Briefly explain Five-level schema architecture to support global applications in the Federal Distributed Database. 4

6. (a) Write the guidelines for using subquery in DBMS.

(b) Consider the following Student:

Std_id	Std_name	Std_email	Std_contact	Std_addr	Std_bg	Std_dept	Std_mar
315121003	Afia Khan	a@gmail.com	01913000000	Dhaka	A +ve	CSE	82
315121004	Rabeya	b@yahoo.com	01721111111	Sonargaon	O +ve	CSE	87
555121001	Siam	c@hub.edu.bd	01821222222	Dinajpur	B +ve	EEE	71
536142004	Sadia	d@gmail.com	01911333333	Narayanganj	O +ve	Econ	72
222121003	Siam	e@hub.edu.bd	01551444444	Munshigonj	O -ve	BBA	66

i) Write the SQL query to select only those students who are neither in the CSE department nor in EEE.

ii) Write the SQL query to insert Siam's std_dob = '05-05-1995' in the EEE department.

iii) Write the SQL query to show all data who obtained marks between 70 and 80.

7. (a) Write short notes on transitive dependency and loss less join.

(b) Consider the following **Student** table with variable repeating groups. Find 1NF, 2NF and 3NF for the given order table.

Student

Std_Id	Adm_Date	Std_name	Std_email	Grade	Cr_Points	Course_Name	Ob_Mark
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Discrete	76
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Algorithms	77
31501	11/03/18	Dev Joe	dev@iewr.com	A	1	Fundamentals	76
31502	12/03/18	Dany Sam	dany@ter.com	B	4	Algorithms	66
31502	12/03/18	Dany Sam	dany@ter.com	B	4	Fundamentals	67



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE315: Microprocessors & Microcontrollers

Batch: 23th

Date: 19/03/2025

Final Examination

Semester: Fall 2024

Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the following.

5 × 8 = 40

1. (a) Define coprocessor. Explain the role of a math coprocessor in a computer system. 3
(b) Describe the advantages of using a math coprocessor compared to using a general-purpose microprocessor like the 8086 for mathematical operations. 3
(c) What are the different data types handled by the 8087 coprocessor? Provide a brief description of each. 2
2. (a) Describe the function of the 8087 register stack and the role of the Stack Pointer (SP). How does data get pushed and popped from the stack? 4
(b) What is the significance of the Status Register and the Control Register in the 8087 coprocessor? 4
3. (a) Explain the function of the 8254 Programmable Interval Timer. Why is it essential in microcomputer systems? 2
(b) Explain the role of the 8254 in generating accurate time delays. Why is hardware delay considered more efficient than software delay? 2
(c) Illustrate the internal block diagram of the 8254. What are the major components involved in its operation? 4
4. (a) Briefly explain the different modes of operation in the 8254. 3
(b) For Counter 2 of Intel 8254, generate the Control Word and illustrate the timing diagram for a binary count value of 6 in rate generator mode using 16-bit write operation. 5
5. (a) What are the two primary reasons for managing memory in a multitasking operating system? 2
(b) Explain the concept of overlays in memory management. How overlays are used to manage limited memory in single-user systems? Explain with illustration. 3
(c) What are the two main methods for managing the available memory of a system? 3

6. (a) Illustrate the architecture of 8051 microcontroller. Highlight each component with brief description. 4
- (b) Why even memory addresses take 1 cycle to access, while odd memory addresses take 2 cycles? 4
7. (a) What are the interrupts in the 8051 microcontroller? 2
- (b) Explain the significance of the reset pin (RST) in the 8051 microcontroller. 2
- (c) Write the purpose and operation of the external oscillator pins in the 8051 microcontroller. How these pins are connected to an external crystal oscillator? 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 318: Computer Networks Lab

Semester: Fall 2024

Time: 1 Hour 30 Minutes

Total Marks: 40

Date: 01/03/2025

Lab Final Question

Part A: 20 Marks

Name: Tamvin Hassan.

Task 1: IP address subnetting using VLSM

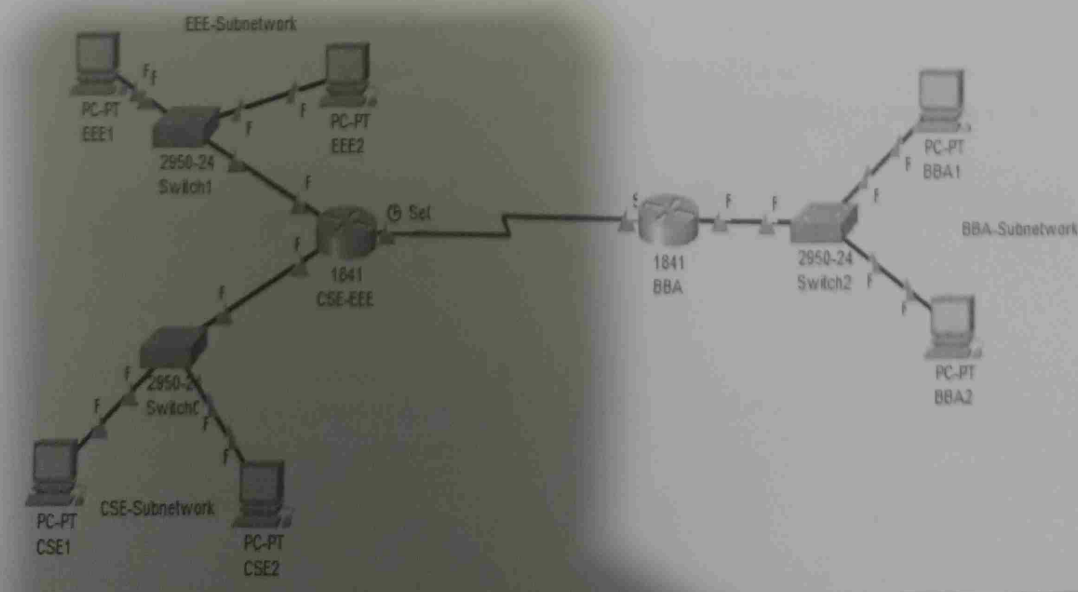
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Consider the following block of IPv4 addresses are given to you: 192.16.0.0/20.

The requirement of IP addresses for different departments are given below:

Department	Required IP addresses
CSE	1000
EEE	200
ME	60
CHEM	60
MATH	20
LINK1	2
LINK2	2
LINK3	2

First create subnets for the different departments according to the requirement given in the above table. Find the IP address assignments to different departments/subnets.



Task 2 (Configure IP address, subnet mask, etc.)

Configure all PCs and routers with IP address, subnet mask, and default gateway information.

IP address assignments rules are following:

- Give CSE1 and CSE2, 3rd and 4th usable IP addresses of its subnetwork
- Give EEE1 and EEE2, 3rd and 4th usable IP address of its subnetwork
- Give BBA1 and BBA2, 1st and 2nd usable IP addresses of its subnetwork
- For router interfaces in CSE, EEE and BBA subnetworks, use the last usable IP address of the respective LANs
- For the WAN of CSE-EEE and BBA router, use the LINK2 block. Assign 1st IP to BBA router and last IP to CSE-EEE router
- For all switches, assign the 10th usable IP address of the respective LANs to VLAN 1 interface so that the switch can be configured remotely.

Task 3 (Configure router information)

Configure the following router configurations:

- Configure clock rate of 72000 in the CSE-EEE router [*This step is very important, otherwise the connectivity will not work*]

Task 4 (Configure static routing)

- Configure a **default static route** in BBA router that will use CSE-EEE router interface as its next hop address.
(The routing entry command should be given from the global configuration mode.)
- Configure a static route in CSE-EEE router.

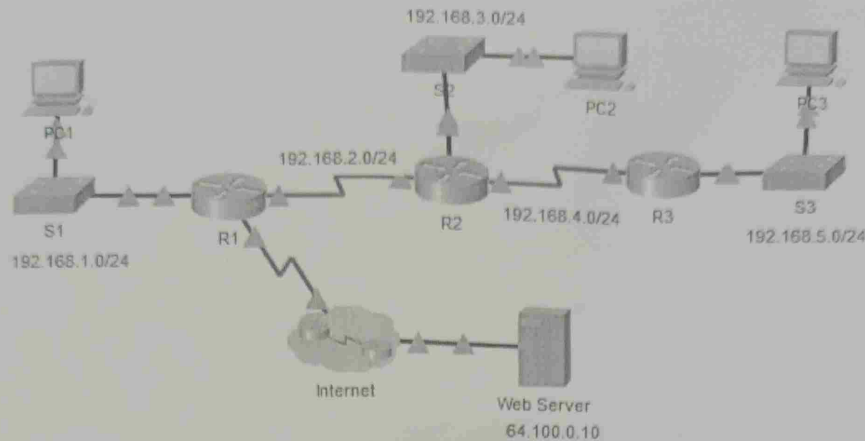
Task 5 (Test your network)

- Ping from BBA1 to CSE1 Pc. Verify that ping is successful.
- Ping from BBA1 to EEE2 Pc. Verify that ping is successful.
- Ping from EEE2 to EEE1 Pc. Verify that ping is successful.

Part B: 20 Marks

Q1: Configure Routing Information Protocol (RIPv2)

10 Marks

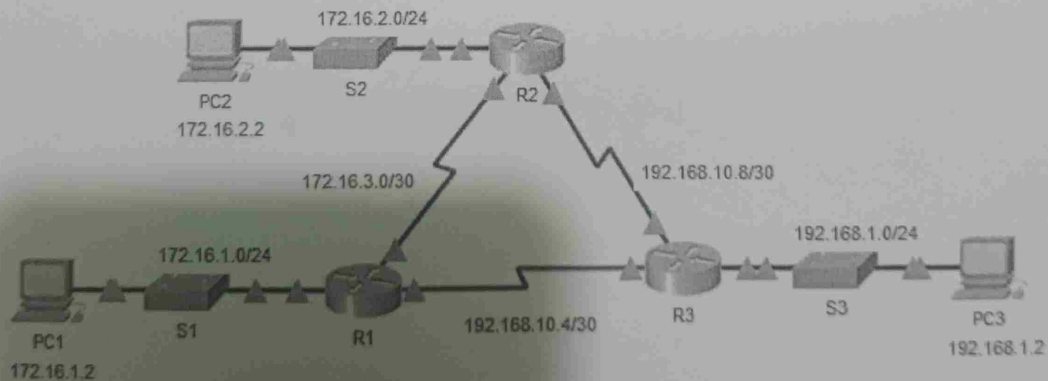


Implement Routing Information Protocol (V2) for above networks. Configure the LAN port that contains no routers so that it does not send out any routing information. Every PCs should ping with each other.

Q2: Open Shortest Path First (OSPFv2)

10 Marks

Write the steps using the following requirements to configure OSPF routing on all three routers:



- Process ID for each router: R1: 11, R2: 12, R3: 13
- Router ID for each router: R1 = 1.1.1.1; R2 = 2.2.2.2; R3 = 3.3.3.3
- Network address for each interface
- LAN interface set to passive



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE316: Microprocessors & Microcontrollers

Batch: 23th

Date: 02/03/2025

Lab Final Examination

Semester: Fall 2024

Time: 40 minutes

Total Marks: 40

Answer All questions of the following.

1. Multiple Choice Questions.

5

- i) Which interrupt is used to terminate a program in 8086 assembly language?
a) INT 20H b) INT 21H c) INT 22H d) INT 23H
- ii) In 8086 assembly language, which register pair is used by default for division operation?
a) AX:BX b) DX:AX c) CX:DX d) BX:CX
- iii) Which of the following assembler directive is used to declare initialized data?
a) .STACK b) .MODEL c) .CODE d) .DATA
- iv) How is a character converted from lowercase to uppercase in 8086 assembly?
a) ADD AL, 20H b) SUB AL, 20H c) ADD AL, 32 d) SUB AL, 32
- v) When multiplying two 8-bit numbers using MUL instruction, where is the result stored?
a) AL b) AX c) BX d) DX

2. True/False Questions. (Explain the false statement)

5

- i) The LOOP instruction automatically decrements the BX register.
- ii) The directive .MODEL determines the memory addressing capabilities of a program.
- iii) The AX register must be used to initialize the data segment register DS.
- iv) Division by zero is allowed in 8086 assembly language.
- v) The ADD instruction affects the flags in the flag register.

3. Fill in the Blanks.

10

- i) The directive _____ is used to mark the end of an assembly language program.
- ii) To convert ASCII code to its numeric value in assembly language, we subtract _____ from it.
- iii) The instruction _____ is used to load the address of a variable into a register.
- iv) When displaying a string in assembly language, the string must end with the _____ character.
- v) The _____ instruction decrements CX register and jumps to a label if CX is not zero.

4. Explain how array elements are accessed in 8086 assembly language.

5

5. Describe the process of converting an input character to its numeric value in assembly language.

5

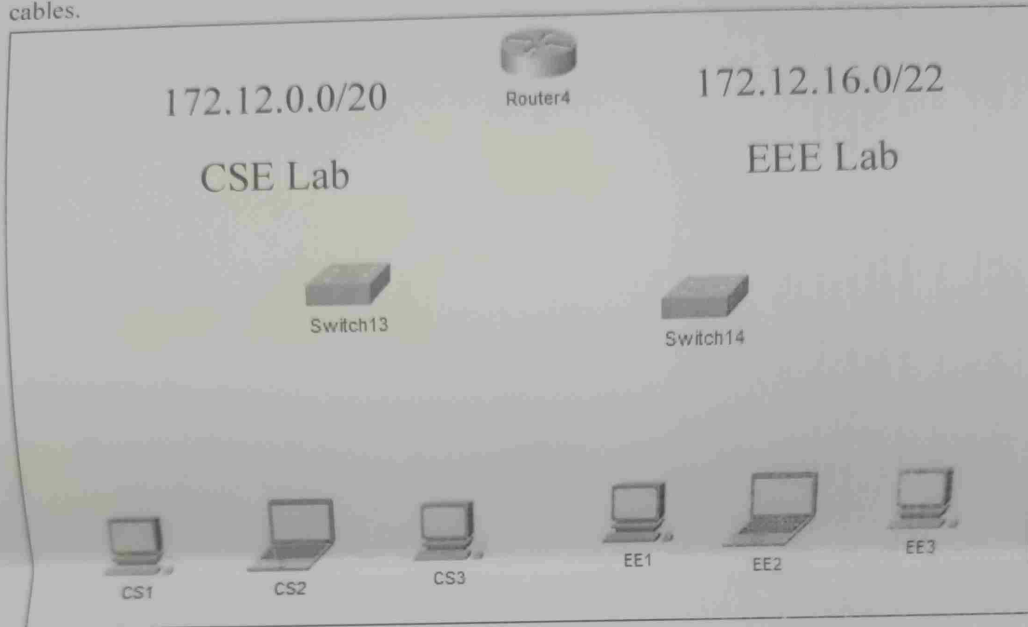
6. Write an assembly language program to convert a lowercase string to uppercase and display it.

10

Computer Networks Lab Assessment

Marks: 30

1. Configure the following network with the following devices and connect with proper cables.



Consider you are given an IP block **172.12.0.0/20** for CSE lab and **172.12.16.0/22** for EEE lab. Configure all PCs and routers with IP address, subnet mask, and default gateway information.

IP address assignments rules are following:

- Give CS1, CS2 and CS3 its 1st, 3rd and 5th usable IP addresses of its subnetwork
- Give EE1, EE2 and EE3 its 2nd, 4th and 6th usable IP addresses of its subnetwork
- For router interfaces in CSE and EEE subnetworks, use the last usable IP address of the respective LANs
- For all switches, assign the 10th usable ip address of the respective LANs to VLAN 1 interface
- Configure hostname for router as 'HUB'
- Configure for the router console and execution password as 'cisco'
- Configure secret password as 'cena'
- Configure banners for both routers as 'This router requires authentic access'
- Encrypt all passwords in both routers
- Save your running configuration into startup configuration



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE316: Microprocessors & Microcontrollers

Batch: 23th

Date: 02/03/2025

Practical Implementation
Tasks

Semester: Fall 2024

Time: 40 minutes

Total Marks: 20

Instructions: Choose (YOUR_ROLL%5 + 1)th of the following tasks and implement it using EMU8086. Your solution will be evaluated based on correctness, efficiency, program structure and output.

1. **Sum of Array Elements**

Create an assembly program that initializes an array of 5 bytes with values of your choice. Calculate the sum of all elements in the array and display the result as a decimal number.

2. **String Reversal**

Write an assembly program that asks the user to enter a string (maximum 10 characters). Display the input string and then display the same string in reverse order.

3. **Simple Interest Calculator**

Create an assembly program to calculate simple interest. Ask the user to input principal (P), rate (R), and time (T) as single-digit numbers. Calculate interest using the formula $I = (P \times R \times T) / 100$ using multiplication and division operations.

4. **Average Calculator**

Write an assembly program that takes 5 single-digit numbers as input. Calculate their sum and then find the average by dividing the sum by 5 using the DIV instruction. Display both the sum and the average.

5. **Division with Error Handling**

Create an assembly program that performs division between two user-input numbers. Implement error handling for division by zero and display appropriate error messages when needed.